Handout

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1. A sample Tolerance Argument (with big conjunctions)

Necessitated Tolerance: Necessarily (if the Great Pyramid is 400 meters tall, then it could have been 399 meters tall; and if it is 399 meters tall, then it could have been 398 meters tall; and … ; and if it is 2 meters tall, then it could have been 1 meter tall).

Iteration: If it is possible for it to be possible for something to be the case, it is possible for it to be the case.

Hypertolerance: If the Great Pyramid could have been 400 meters tall, then it could have been 1 meter tall.

2. Why it's valid (in standard modal logic).

(i) We can get from $\Box(P_1 \wedge \cdots \wedge P_n)$ to $\Box P_1 \wedge \cdots \wedge P_n$

Stepwise Necessitated Tolerance: Necessarily (if the Great Pyramid is 400 meters tall, then it could have been 399 meters tall); and necessarily (if it is 399 meters tall, then it could have been 398 meters tall); and … ; and necessarily (if it is 2 meters tall, then it could have been 1 meter tall).

(ii) We can get from $\Box(P \rightarrow Q)$ to $\Diamond P \rightarrow \Diamond Q$; and hence from $\Box(P \rightarrow \Diamond Q)$ to $\Diamond P \rightarrow \Diamond\Diamond Q$. So we have

Double Possibility Transfer: If the Great Pyramid could have been 400 meters tall, then it could have been such that it could have been 399 meters tall; and if it could have been 399 meters tall, then it could have been such that it could have been 398 meters tal); and … ; and if it could have been 2 meters tall, then it could have been such that it could have been 1 meter tall.

(iii) We can get from Iteration and any $\Diamond\Diamond P$ to $\Diamond P$, and hence from $\Diamond P \rightarrow \Diamond\Diamond Q$ to $\Diamond P \rightarrow \Diamond Q$:

Possibility Transfer: If the Great Pyramid could have been 400 meters tall, then it could have been 399 meters tall; and if it could have been 399 meters tall, then it could have been 398 meters tal); and … ; and if it could have been 2 meters tall, then it could have been 1 meter tall.

(iv) We can get from $P_1 \rightarrow P_2$ and $P_2 \rightarrow P_3$ to $P_1 \rightarrow P_3$; and hence from $P_1 \rightarrow P_2$ and … and $P_{n-1} \rightarrow P_n$ to $P_1 \rightarrow P_n$, and from Possibility Transfer to Hypertolerance.

3. More examples

Bucket argument

Chander, Salmon: originating matter

Chisholm: Adam/Noah.

4. Quick survey of response-strategies, and the challenges they face

(i) Accepting Hypertolerance (Penelope Mackie…)

- Chisholm seems to suggest an argument against Hypertolerance (in his Adam/Noah argument) from Anti-Haecceitism.
- Salmon, following Kripke footnote 56, develops a detailed argument against Hypertolerance (as regards originating matter) from *sufficiency-of-origins* principles along the lines of (but more cautious than) the following:

  > *Simple Sufficiency of Origins*: If table x originates from hunk h, then necessarily any table that originates from h is x.

- One can also make arguments against Hypertolerance (in certain "super-fine-grained" Tolerance Arguments) from Microphysical Supervenience.
- And there is a worry that the strategy of accepting Hypertolerance doesn't give us a good response to Robustness Arguments, where the premise is not just that it's *possible* for the object to have a certain slightly different property, but hard for there to be something other than that object that has that property.

(ii) Denying Iteration (Hugh Chandler, Nathan Salmon, David Lewis, Teresa Robertson…)

- What if we define a notion of 'ancestral necessity' for which Iteration holds automatically? We'll be driven to accept Hypertolerance for it: is that OK?
- Does Iteration denial conflict with the characterisation of metaphysical necessity as 'absolute'?
- Is Iteration-denial strategy plausible for narrower modalities like *having positive objective chance* (at such and such time)?

(iii) Denying Necessitated Tolerance and even plain Tolerance

- This seems to lead to a radical error theory about giant swathes of our ordinary modal judgments.
- NB these motivations have nothing to do with the appeal of the Sorites premise.

(iv) Denying Necessitated Tolerance while keeping plain Tolerance

- One could try to make something of the principle that Tolerance is known a priori together with the idea that a priority requires necessity; but that's not so promising.
- A widespread refrain is that this option makes the actual world 'special' in an objectionable way; but this is to pin down.
- We think the big obstacle here is the "Security Argument".

5. Replacing big conjunctions with quantifiers

When presented with an argument with a large number of premises motivated in similar ways—or a very complex premise featuring a big conjunction—it's often wise to try to articulate the more general thought that lies behind them, which will normally involve some kind of quantification.

Introduce a *closeness relation* ~ among properties.

- For example: F~G could mean: 'There are collections of atoms C and D that overlap by at least 90% and contain equally many atoms with each atomic number, such that F = *being originally composed of* C and G = *being originally composed of D.*

Define *ancestral closeness* ~*. Intuitively, F~*G means that for some n, and some sequence $H_1 \ldots H_n$, F~H_1 and H_1~H_2 and … and H_n~G. We can say this in higher order logic without bringing in numbers and sequences.

- Of we can just choose some specific large n and work with $\sim^n$ instead of ~*.

> *Necessitated Tolerance:* $\Box\forall F\forall G(F\sim G \rightarrow (Fa \rightarrow \Diamond Ga))$
>
> *Persistent Closeness:* $\forall F\forall G(F\sim G \rightarrow \Box F\sim G)$
>
> *Iteration:* $\forall p(\Diamond\Diamond p \rightarrow \Diamond p)$
>
> ————
>
> *Hypertolerance*: $\forall F\forall G(F\sim^* G \rightarrow (\Diamond Fa \rightarrow \Diamond Ga))$

Problem: in many instantiations. Persistent Closeness isn't very plausible if □ means metaphysical necessity.

A possible solution: define a narrower □ that "holds fixed" the property-closeness facts, making Persistent Closeness automatic, without doing much to shake the plausibility of Necessitated Tolerance.

6. Quantified Tolerance Arguments

The arguments up to now have all required picking a particular object *a* to argue about. But obviously in many cases the claims about *a* are motivated by generalizations more general category to which *a* belongs, e.g. tables, or wooden tables… Again, it's a good idea to try to capture this explicitly.

There are actually two different ways of turning the above argument form using singular terms into an argument for a quantified Hypertolerance claim:

Quantified Hypertolerance: $\forall x(Kx \to \forall F\forall G\forall x(F{\sim}^{*}G \to (\Diamond Fx \to \Diamond Gx)))$

The less interesting way uses

Wide Scope Necessitated Tolerance: $\forall x(Kx \to \Box\forall F\forall G(F{\sim}G \to (Fx \to \Diamond Gx)))$

The more interesting way uses

Narrow Scope Necessitated Tolerance: $\Box\forall F\forall G\forall x(F{\sim}G \to ((Kx \wedge Fx) \to \Diamond(Kx \wedge Gx)))$